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Tuning the Cross-Linker Crystallinity of a Stretchable Polymer Semiconductor

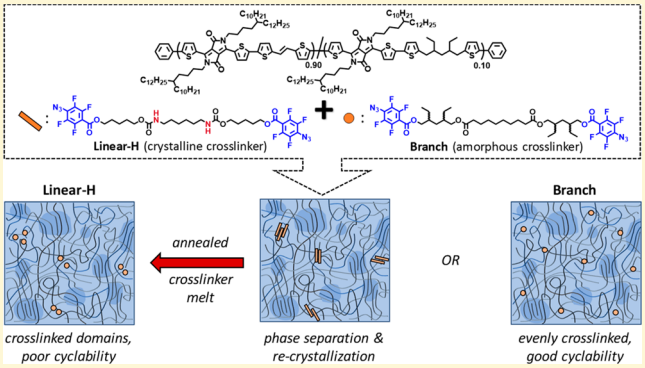
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Supporting Information

ABSTRACT: The cross-linking of conjugated polymers has been demonstrated to be an effective strategy to improve its elastic properties to give deformable semiconductors for plastic electronics. While there have been extensive studies of the structural requirements of the polymer host for good film ductility, no work to date has focused on the relevance of the structural design or chemistry of these cross-linker additives. In this study, urethane groups and tertiary carbon atoms are inserted into the alkyl backbone of perfluorophenyl azide-based cross-linkers to investigate the importance of cross-linker crystallinity with respect to polymer morphology and hence mechanical and electrical properties. Linear cross-linkers with hydrogen bonding from urethane groups readily phase separate and recrystallize in the polymer network to form cross-linked domains that obstruct the strain distribution of the polymer film. Branch cross-linkers with tertiary carbon on the other hand form an evenly cross-linked network in the polymer blend stemming from excellent miscibility and show a 4-fold increase in fracture strain. Furthermore, a stable hole mobility of 0.2 $cm^2\ V^{-1}\ s^{-1}$ is achieved up to ε = 100%, and a stable hole mobility of 0.1 $cm^2\ V^{-1}\ s^{-1}$ after 2000 cycles of ε = 25% on fully stretchable organic field-effect transistors.



1. INTRODUCTION

The evolution of intrinsically stretchable polymer semiconductors for wearable and implantable electronics has gained much traction in the past few years.[1−7] Ductile semiconductive films[8,9] have been achieved, as have biodegradable,[10−12] healable,[13,14] and solution processable[15,16] materials that are suitable for low-cost plastic bioelectronics. Thus far, much attention has been paid to better understanding the structural requirements of polymer semiconductors such as side chains and the backbone for next-generation deformable electronics.[17−20] Introducing branching into alkyl side chains and inserting individual aromatic rings into polymer backbones were shown to improve backbone flexibility and decrease the overall crystallinity relative to those of polymers with linear side chains and extended fused rings.[21,22] Similarly, incorporation of soft nonconjugated segments could increase the deformability by phase separation[8,9] or introduce additional functionalities.[13,23] These studies have provided the basis for our current understanding of morphological requirements for ductile electronics.

Aside from structural alterations, others have focused on postpolymerization modification strategies to optimize the mechanics of known polymer semiconductors. These methods include physical blending of elastomers[24−28] and small additives[29,30] as well as chemical cross-linking.[31] In our previous study, we developed a stretchable diketopyrrolopyrrole (DPP)-based polymer system through covalently cross-linking the polymer with oligo-siloxane.[31] The cross-linked film displayed an increase in yield strain along with fatigue resistance. Contrary to our expectations, a decrease in the elastic modulus and an increase in fracture strain were also observed because of a plasticizing effect from the soft siloxane cross-linker.

Here, we investigate the importance of cross-linker "hardness" with respect to the mechanical properties of cross-linked polymer semiconductors through systematic tuning of the cross-linker's crystallinity. While the literature on the effects of polymer crystallinity on its charge transport and mechanical properties is extensive, this study presents a first look at the influence of additive crystallinity on the mechanics of cross-linked semiconductors. This is done by synthesizing a series of four alkyl cross-linkers with various amounts of branching and

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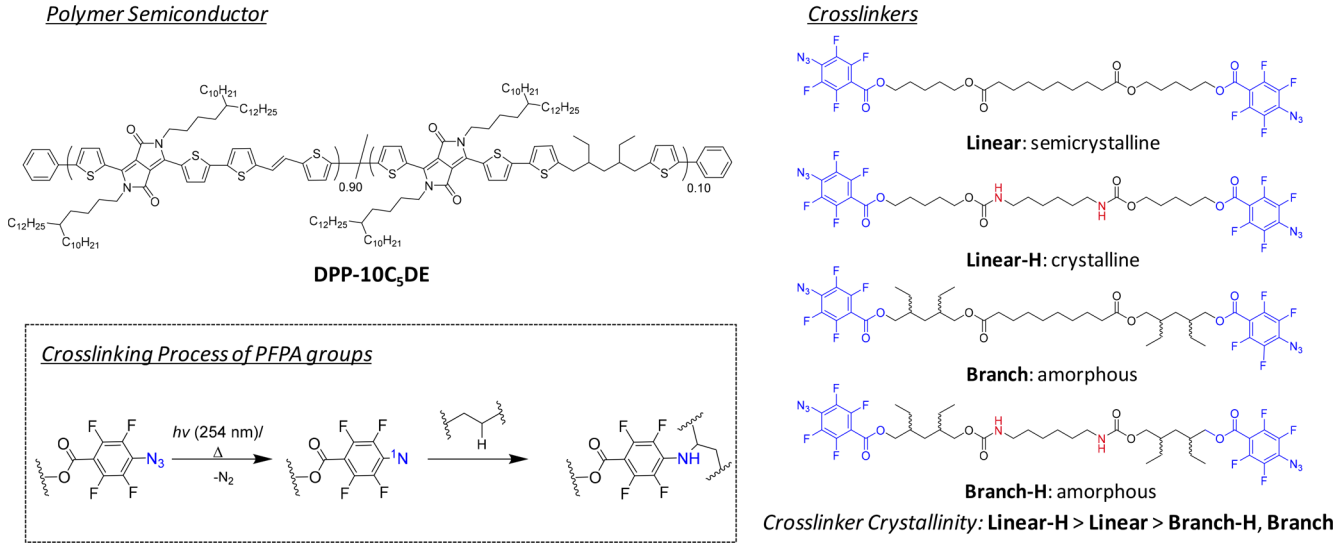
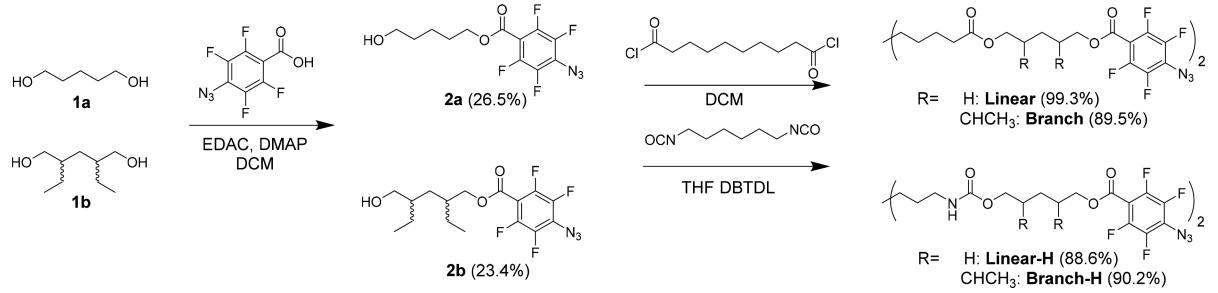


Figure 1. Chemical structure of the four cross-linkers and the DPP-based polymer used in this study. The crystallinity of the cross-linkers increases with the number of urethane groups and decreases upon the introduction of branching. The simplified scheme of the perfluorophenyl azide (PFPA) cross-linking reaction is displayed.

Scheme 1. Synthetic Pathway toward the Different Cross-Linkers



urethane linkages to disrupt and promote crystallization, respectively (Figure 1). An sp^3 carbon backbone is selected for its excellent miscibility with the alkyl side chains of conjugated polymers. Previously, the siloxane cross-linker and DPP polymer mixture had to be partially cross-linked in solution prior to deposition because of its poor wettability stemming from the poor miscibility between the cross-linker and polymer network.[31] Urethane groups are added in this study to introduce hydrogen bonding (H-bonding) units to encourage self-assembly[32,33] and potentially provide an energy-dissipative mechanism for better stretchability.[13] On the other hand, tertiary carbons with ethyl groups are added to introduce "kinks" and steric hindrance to the cross-linker, which suppresses crystallization and aggregation.[34]

In addition, the cross-linkers are end-capped with perfluorophenyl azide (PFPA) groups, which releases nitrogen gas and forms a singlet nitrene upon thermal activation or photoactivation. The singlet nitrene then attacks available sp^3 C−H bonds on the alkyl side chains to form covalent cross-links (Figure 1).[35] As the cross-linking reaction is catalyst-free and nonspecific relative to our previous system that required alkene groups and a platinum catalyst for cross-linking by hydrosilylation, the PFPAs are "universal" cross-linkers that do not require custom-made polymers with specific functional groups. Furthermore, PFPA cross-linkers have been demonstrated to efficiently cross-link and photopattern conjugated polymers without significant degradation of their electronic properties.[36]

However, the use of PFPA cross-linkers to modify the mechanical properties of polymer networks has not yet been investigated. We anticipate cross-linker crystallinity to have a profound impact on film morphology and hence the deformability and optoelectronic properties of polymer semiconductors. Lastly, **DPP-10C$_5$DE** (Figure 1) was selected as the polymer semiconductor used in this study and synthesized following a previously described procedure.[34] It was selected for its long alkyl side chains and flexible linker in the backbone, which ensures a good solubility and a high density of active sites for efficient cross-linking while maintaining excellent charge transport properties.

2. RESULTS AND DISCUSSION

To examine the effects of branching and H-bonding on cross-linker crystallinity, a linear alkyl cross-linker with ester linkages was synthesized as a control. The synthetic route is presented in Scheme 1. A similar PFPA cross-linker with a different alkyl length was previously synthesized by our group and used in the fabrication of a stretchable transistor array.[37] Commercial 1,5-pentanediol (**1a**) was first monofunctionalized with 4-azido-2,3,5,6-tetrafluorobenzoic acid by Steglich esterification to give **2a**, which was subsequently reacted with sebacoyl chloride to yield our control **Linear** cross-linker. Hexamethylene diisocyanate was reacted with **2a** in place of sebacoyl chloride to give the **Linear-H** cross-linker that is as long as our control cross-linker but contains urethane linkages as opposed to ester

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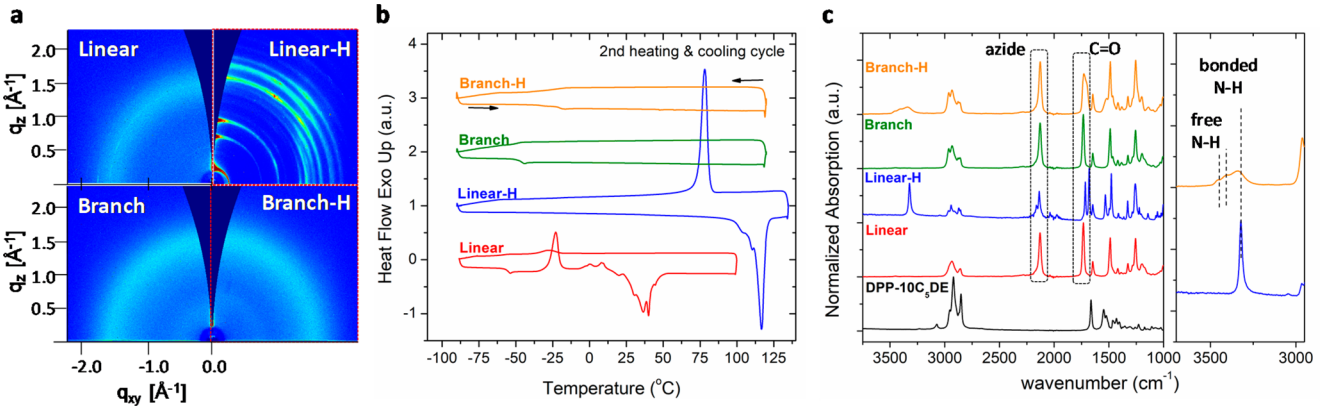


Figure 2. Evaluation of cross-linker crystallinity. (a) Two-dimensional grazing-incidence X-ray diffraction (GIXD) patterns of cross-linkers. The cross-linkers were dissolved in chlorobenzene at a concentration of 25 mg mL^{-1} and spun on Si substrates for measurements. All images were scaled for exposure time and illuminated volume for qualitative comparison across samples. (b) Differential scanning calorimetry (DSC) heating and cooling traces of the cross-linkers. (c) Attenuated total-reflectance Fourier transform infrared (ATR-FTIR) spectroscopy of the neat polymer and cross-linkers. The right spectra show N−H stretching from urethane groups in **Linear-H** and **Branch-H**.

groups. To introduce branching without altering the length of the cross-linkers, **1a** was replaced with 2,4-diethyl-1,5-pentanediol (**1b**) to give cross-linkers **Branch** and **Branch-H**. **1b** contains two chiral centers and is a mixture of *S-S*, *S-R*, and *R-R* isomers that was used previously to produce benign solvent processable conjugated polymers.[34] The detailed cross-linker synthesis and purification procedures are available in the Supporting Information.

2.1. Cross-Linker Characterization. The physical appearances of the cross-linkers were notably different. **Linear** was a waxy solid, and **Linear-H** was a solid powder; **Branch** and **Branch-H** were viscous oils (Figure S6). To quantitatively determine the crystallinity of the cross-linkers, differential scanning calorimetry (DSC) and grazing-incidence X-ray diffraction (GIXD) were performed. Both **Linear** and **Linear-H** exhibited an intense melting peak in their DSC heating traces at 40 and 117 °C, respectively, indicating crystalline packing in the bulk material at room temperature (Figure 2b). **Linear-H** had a higher melting temperature (T_m) and a higher enthalpy of fusion (ΔH_{fus}), suggesting larger crystalline domains and a higher degree of crystallinity due to H-bonding from the urethane groups (Table 1). **Branch** and **Branch-H** on the other hand appeared to be fully amorphous with no T_m; glass transition temperatures (T_g) from the amorphous chains were recorded at −44 and −22 °C, respectively. **Branch-H** required more energy for molecular motion (higher T_g) because of the additional noncovalent interactions. Notably, **Linear** also had a T_g of −57 °C, revealing the cross-linker's semicrystalline nature with both crystalline and amorphous fractions. A cold crystallization peak was also observed at −23 °C despite a cooling rate of 5 °C min^{-1}, implying the longer time scale required for crystallization relative to **Linear-H**. Similar conclusions could be drawn from the GIXD measurements of the cross-linkers in thin films. As shown in Figure 2a, **Linear-H** is highly crystalline with clear diffraction patterns. **Linear** displayed some weak diffraction peaks, indicating a low level of crystallinity, whereas the branch cross-linkers were completely amorphous with no observable diffraction patterns.

Table 1. Physical Properties of Cross-Linkers

cross-linker	T_g (°C)[a]	T_m (°C)[b]	ΔH_{fus} (J g^{-1})[c]	T_a (°C)[d]	T_d (°C)[e]
Linear	−56.9	40.0	60.3	169.8	390.9
Linear-H	–	116.5	91.1	174.2	368.6
Branch	−44.4	–	–	173.6	394.7
Branch-H	−21.9	–	–	172.6	361.5

[a]Glass transition temperature. [b]Melting temperature. [c]Enthalpy of fusion. [d]Activation temperature for PFPA as measured by thermogravimetric analysis at a rate of 10 °C min^{-1}. [e]Decomposition temperature.

The thermal stability of the cross-linkers was studied by thermogravimetric analysis (TGA) (Figure S7), and the results are summarized in Table 1. An initial decrease in mass was first observed around 172 °C, which was the activation temperature of the azide cross-linkers to form singlet nitrene and release nitrogen gas. Decomposition was observed above 350 °C, indicating the good thermal stability of the linkages in all cross-linkers.

To further investigate the nature of H-bonding in the urethane groups of the cross-linkers, attenuated total-reflectance Fourier transformation infrared spectroscopy (ATR-FTIR) was performed. All cross-linkers exhibited a strong azide stretch around 2150 cm^{-1} and a carbonyl (C═O) stretch around 1750 cm^{-1}, which are distinctive from semiconductor **DPP-10C$_5$DE** (Figure 2c). Furthermore, both **Linear-H** and **Branch-H** showed characteristic N−H stretching in the region of 3300−3500 cm^{-1} from the amide groups. **Linear-H** had a sharp N−H peak around 3350 cm^{-1} suggesting the occurrence of N−H···O═C H-bonding, whereas **Branch-H** displayed both bonded and free N−H peaks around 3350 and 3450 cm^{-1}, respectively, indicating partial formation of H-bonding between cross-linkers (Figure 2c).[38] Notably, the N−H stretch is sharp for **Linear-H** in comparison to the broad peaks of **Branch-H**, revealing the respective crystalline and amorphous nature of the H-bonded N−H. This implies that the amide groups in **Branch-H** could overcome the sterics introduced through branching to still form H-bonds.

2.2. Cross-Linked Film Characterization. The cross-linked films were prepared by mixing the neat polymer semiconductor solution (10 mg mL^{-1}) with 5 wt % cross-linker (25 mg mL^{-1}) and spin-coating on a SiO_2 substrate or glass to give 50 nm thin films. The films were then cured at 150 °C for 30 min under nitrogen to give **5Linear**, **5Linear-H**, **5Branch**,

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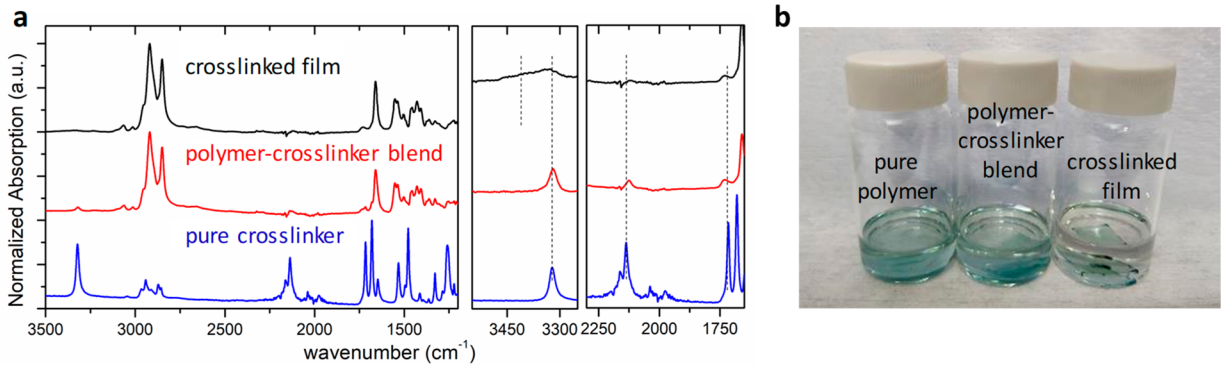


Figure 3. Evaluation of cross-linking efficiency. (a) Attenuated total-reflectance Fourier transform infrared (ATR-FTIR) spectroscopy of pristine and cured polymer-cross-linker films with 5 wt % **Linear-H**. The middle spectra show N−H stretching from urethane groups. The right spectra show azide and carbonyl peaks from the cross-linker. (b) Photograph of neat semiconductor, pristine **5Linear-H**, and cured **5Linear-H** (from left to right, respectively) thin films soaked in chloroform. **5Linear-H** remains insoluble after thermal cross-linking.

and **5Branch-H**. Despite yielding an activation temperature of ∼172 °C by TGA for the bulk cross-linkers, ATR-FTIR revealed that 150 °C was sufficient to fully cure the thin films. As shown in Figure 3a, the pristine film of the polymer-**Linear-H** blend (red spectrum) showed clear peaks at 2150 and 1750 cm^{-1} corresponding to the cross-linker's azide and carbonyl stretch, respectively. However, after curing (black spectrum), the azide peak completely disappeared whereas the carbonyl peak remained present, indicating that the cross-linker was still in the film but its azide groups were fully activated. The efficiency of the cross-linking was subsequently tested by soaking the cross-linked films in chloroform, which were insoluble (Figure 3b). On the other hand, the neat polymer and pristine polymer-**Linear-H** blend instantaneously dissolved once immersed in chloroform to give a light green solution. In addition, after curing, the N−H peak of cross-linked **5Linear-H** broadens and exhibits both H-bonded and free N−H stretch (Figure 3a) similar to that previously observed for **Branch-H**. As the curing temperature is above the T_m of **Linear-H**, the crystalline cross-linker melted prior to cross-linking but could not recrystallize once it was cooled back to room temperature due to the constraints of the newly formed covalent bonds. This resulted in an amorphous cross-linked network like the other cross-linkers.

2.2.1. Cross-Linked Film Morphology. A good grasp of the cross-linker additive's impact on film morphology is crucial for the understanding of their mechanical and electrical behavior. UV−vis analysis of the polymer-cross-linker mixture did not exhibit any observable differences in the solution (Figure S8) or in the film state, indicating no electronic interactions between the two species, yet upon annealing, a significant reduction in the 0−0 vibronic transition near 800 nm was detected for the cross-linked films, indicating a decrease in the level of polymer aggregation (Figure 4). A similar trend was observed for the crystalline region of the polymer films based on GIXD. As shown in Figure 5a, a weakened out-of-plane (h00) alkyl diffraction and (010) $\pi-\pi$ stacking peak was observed for the cross-linked film, revealing a decrease in polymer crystallinity. The meridian line-cut plots (Figure 5b) further indicate a lower level of crystallinity for the branched cross-linkers as well as a slight shift in peak position for all cross-linked films. Table 2 presents the calculated crystallographic parameters of the cross-linked films, which showed an increase of ∼0.6 Å in lamellar spacing for all cross-linked films relative to the neat polymer film. On the contrary, a negligible

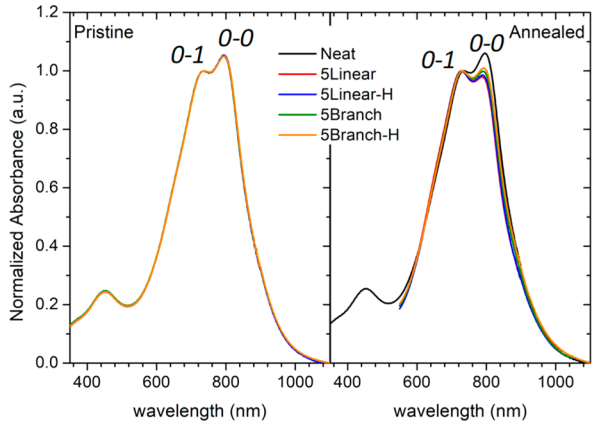


Figure 4. Normalized UV−vis absorption spectra of cross-linked polymer films spin-coated on glass before (left) and after (right) curing.

change in $\pi-\pi$ spacing was recorded, suggesting that the semiconductor crystalline regions were mostly disrupted in the lamellar stacking direction, likely because of the favored interaction between the alkyl side chains and cross-linkers. However, a change of 0.6 Å is small relative to the length and size of the cross-linkers; hence, the insertion of the cross-linkers into the polymer crystalline region is highly unlikely. Instead, the distortion of neighboring amorphous regions by cross-linking may lead to small changes in crystalline packing.

To further investigate whether the cross-linkers were inserted into the polymer crystallites, DSC was performed to check for co-crystallization and evaluate polymer-cross-linker miscibility. In agreement with GIXD, no additional signal was observed for the cross-linked films other than the side-chain T_g from the neat polymer (Figure S9). Interestingly, for the pristine polymer-cross-linker blends, the T_m values of the linear cross-linkers (**Linear** and **Linear-H**) could be observed (Figure 6a). Co-crystallization typically leads to a shift in T_m;[39] however, the T_m values observed for the pristine blends were almost identical to that recorded for the cross-linkers. This suggests that pure cross-linker crystalline domains were spontaneously formed initially in the linear cross-linker blends prior to thermal annealing. The branched cross-linker blends on the other hand did not reveal any additional signal from the cross-linkers, suggesting good miscibility with the semiconductor host. The recrystallization of the pure cross-linker

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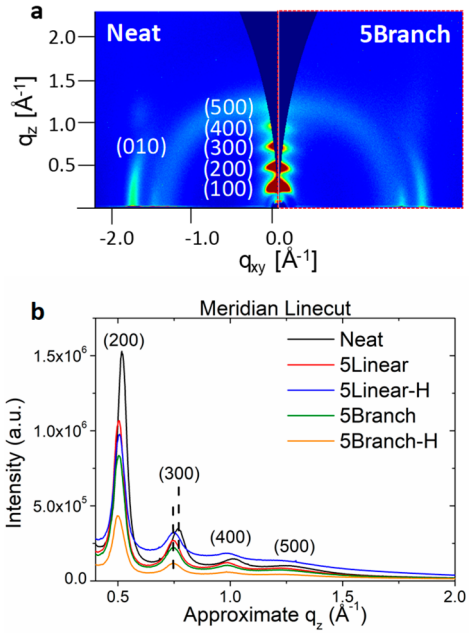


Figure 5. Grazing-incidence X-ray diffraction of the cross-linked polymer. (a) GIXD diffractogram of the neat polymer (left) and **5Branch** (right). (b) Polarization-corrected meridian ($q_{xy} \approx 0$) line cuts of all cross-linked films. Polymer films were spin-coated from a 10 mg mL^{-1} chlorobenzene solution on Si substrates with annealing at 150 °C under nitrogen for 30 min. All images were scaled for exposure time and illuminated volume for qualitative comparison across samples. A diffractogram of the remaining cross-linked films can be found in Figure S12.

Table 2. Crystallographic Parameters of Cross-Linked Polymer Films

	lamellar spacing (Å)[a]	lamellar peak full width at half-maximum (Å^{-1})	$\pi-\pi$ spacing (Å)[b]	$\pi-\pi$ peak full width at half-maximum (Å^{-1})
neat	24.3	0.05	3.5	0.10
5Linear	24.9	0.05	3.6	0.12
5Linear-H	24.8	0.06	3.6	0.11
5Branch	24.9	0.06	3.6	0.13
5Branch-H	25.0	0.06	3.6	0.14

[a]Extracted from fitting the (200) diffraction peak. [b]Extracted by fitting the (010) diffraction peak.

crystalline domain upon film deposition was further confirmed by GIXD of the pristine polymer-**Linear-H** blend, which showed a unique diffraction pattern that originated from the cross-linker (Figure 6b). These features were especially apparent when comparing the meridian line cuts of pure **Linear-H** and the polymer-cross-linker blend (Figure 6c). These features however disappeared upon curing, due to the melting of the cross-linker crystals and formation of amorphous cross-linked network as suggested previously by ATR-FTIR. No cross-linker diffraction pattern was recorded for pristine **5Linear** possibly due to the lower crystallinity of the cross-linker (Figure S11).

The formation of pure cross-linker crystals in pristine **5Linear-H** as revealed by the presence of diffraction and melting similar to those of the pure cross-linker in GIXD and DSC, respectively, implies phase separation in the polymer-cross-linker blend. Optical microscopy and atomic force microscopy (AFM) were performed to confirm this. Indeed, as displayed in Figure 7, clear phase separation and recrystallization of **Linear-H** were observed on the film surface by optical microscopy. Upon thermal curing, smaller domains were formed. By AFM, these cross-linker domains were shown to have shrunk from ∼1 to 0.1 μm in size. On the contrary, smooth films with no observable phase separation were recorded for **5Linear**, **5Branch**, and **5Branch-H** by both optical microscopy and AFM. The surface roughness of the films was also much lower with a maximum of 10 nm in height difference in comparison to that of **5Linear-H**, which exhibits a height difference of 70 nm.

Thus far, we have revealed two vastly different film morphologies upon thermal cross-linking our DPP polymer with the newly synthesized PFPA cross-linkers. The first is from the amorphous cross-linkers **Branch** and **Branch-H**, which demonstrated good miscibility with the polymer host, leading to an evenly cross-linked network in the amorphous region. The second is from the semicrystalline and crystalline cross-linkers **Linear** and **Linear-H**, respectively, which exhibited spontaneous phase separation and recrystallization upon film deposition according to DSC, GIXD, optical microscopy, and AFM measurements. These crystalline domains then melt and form localized cross-linked domains upon thermal annealing as illustrated in Figure 8. The cross-linked domains are likely in an amorphous state, because broadened N−H peaks were observed by ATR-FTIR, and no pure cross-linker diffraction or change in polymer spacing was observed by GIXD. DSC also did not reveal any crystalline peaks or evidence of co-crystallization. Notably, the **Linear** cross-linker recrystallized at a smaller scale due to its semicrystalline nature and could be detected only by DSC. The next step is to investigate how these two different morphologies may lead to different mechanical and electric properties.

2.2.2. Cross-Linked Film Mechanical Properties. The polymer thin films were first transferred to a PDMS elastomer for crack-onset (COS) assessment. As pointed out in previous studies, the monitoring of cracks by optical microscopy at increasing strains is only a qualitative assessment of thin film ductility due to its strong dependence on substrate moduli and surface energies.[2,40] Nonetheless, COS measurement is an effective method for observing trends. The microscope images of the strained polymer films can be found in Figures S14 and S15. Indeed, an increase in COS was observed for all cross-linked films from ε = 25% for the neat polymer film to around ε = 100% for the cross-linked films (Table 3). **5Linear-H** showed the least improvement with a COS of ε = 75%; this was likely due to the rough surface it had and the formation of cross-linked domains, which led to imbalanced stress dissipation across the film and acted as defects.

Pseudo-free-standing tensile test measurements were performed by suspending polymer films on water (FOW) for a more quantitative evaluation of the cross-linked films' mechanics following a previously described procedure.[17,40−43] In agreement with COS measurements, a clear enhancement in polymer ductility was observed upon cross-linking. The stress−strain behavior of the maximum fracture strain obtained for each cross-linker is presented in Figure 9a, and the average elastic modulus and fracture strain are summarized in Table 3. As shown in Figure 9a, the fracture strain of the linear cross-linked films was improved by 2-fold, whereas the branch cross-linkers showed an increase of 4-fold. This may originate from the good miscibility of the branched cross-linkers with the neat

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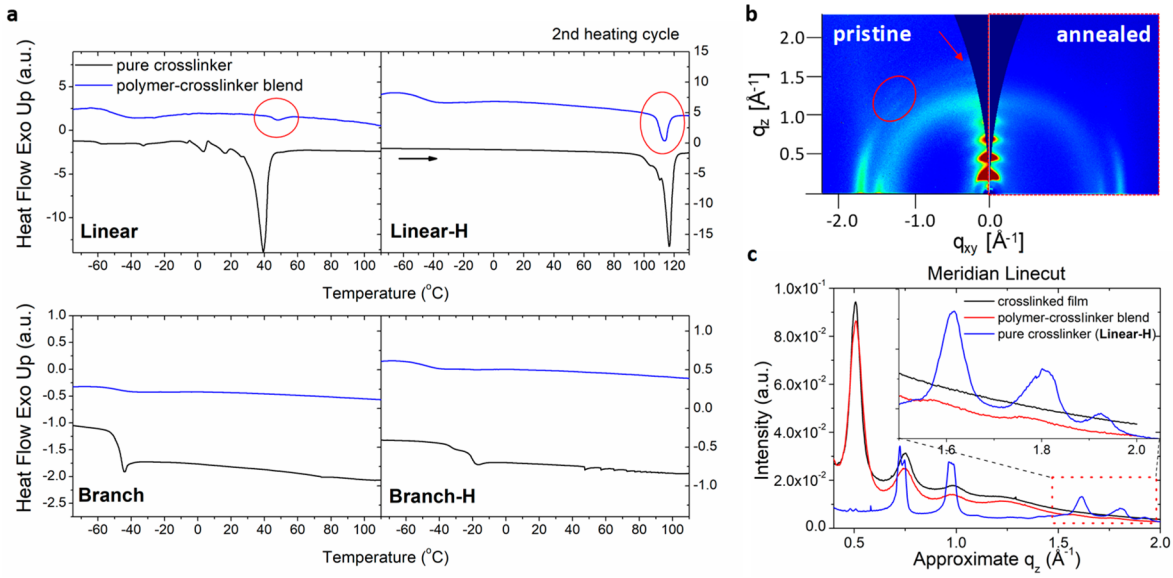


Figure 6. (a) DSC heating traces of pristine polymer-cross-linker blends compared with that of their respective cross-linker. (b) GIXD diffractograms and (c) meridian ($q_{xy} \approx 0$) line cuts of pristine and annealed **5Linear-H** compared with those of **Linear-H**. All images were scaled for exposure time and illuminated volume for qualitative comparison across samples. The red circle and arrow in the diffractogram indicate diffraction features from pure **Linear-H**. The diffractogram of remaining polymer-cross-linker blends can be found in Figure S11.

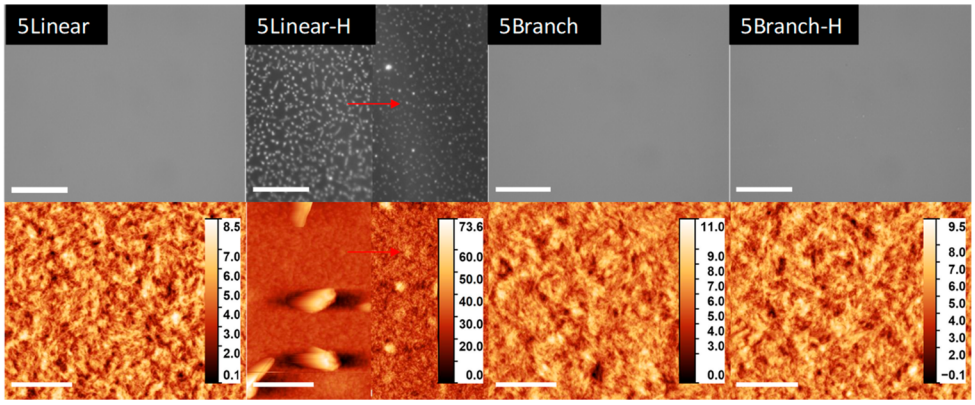


Figure 7. Bright field optical microscope (top) and atom force microscope (bottom) images of cross-linked polymer films. The left panel of **5Linear-H** displays the morphology of the precured film, whereas the right panel shows the cross-linked film morphology. A dark field was applied for the **5Linear-H** microscope image to better reveal the cross-linker domains. Scale bars in microscope images are 20 μm. Scale bars in AFM images are 500 nm, except for that of **5Linear-H**, which is 1 μm.

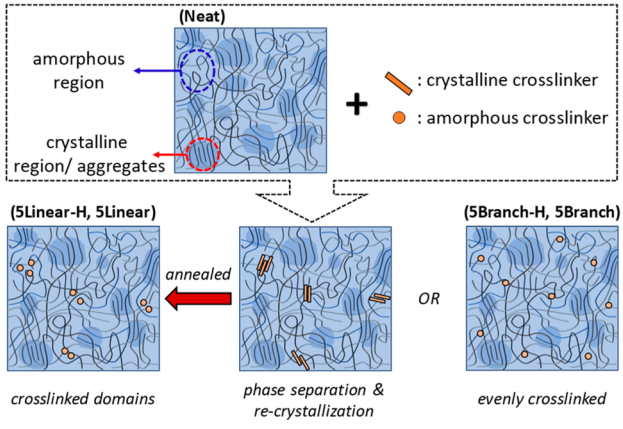


Figure 8. Schematic summarizing the proposed morphological difference between the cross-linked films and the neat polymer film.

polymer that gave a more uniform cross-linked network as discussed above. On the basis of FOW measurements, the presence of amide groups in the cross-linkers did not significantly impact the mechanical behavior of the cross-linked films in comparison to the alkyl branching introduced, as the cross-linkers with H-bonding exhibited mechanical behavior identical to that of their respective cross-linkers with ester linkages.

Additionally, the elastic modulus (E) of each polymer film could also be extrapolated from the FOW measurements. As shown in Table 3, the E values of all of the polymer films were essentially the same (considering the error involved) despite showing a reduction in the level of aggregation and crystallinity by UV−vis and GIXD for the cross-linked films. This is in contrast to our previous study using a siloxane cross-linker that revealed a suppression of E upon cross-linking.[31] Furthermore, no change in E or fracture strain was recorded when comparing the pristine films to each other. This suggests that there was no plasticizing effect from the cross-linker additives. It was only

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Table 3. Mechanical Properties of Polymer Films

	condition	E (MPa)[a]	fracture strain (%)[b]	COS (%)[c]
neat	pristine	225 ± 25	1.2 ± 0.5	25
	annealed[d]	225 ± 20	2.5 ± 1.0	25
5Linear	pristine	205 ± 25	1.3 ± 0.4	50
	annealed	234 ± 30	6.0 ± 1.5	100
5Linear-H	pristine	234 ± 15	1.1 ± 0.7	50
	annealed	230 ± 30	6.0 ± 2.0	75
5Branch	pristine	236 ± 20	2.0 ± 0.6	75
	annealed	204 ± 30	9.0 ± 2.0	100
5Branch-H	pristine	200 ± 25	1.3 ± 0.4	75
	annealed	215 ± 35	8.0 ± 2.0	>100

[a]Average elastic modulus of the polymer thin film as measured by film-on-water experiments. Polymer films were processed to a thickness of ∼50 nm before being further stretched at a strain rate of 5×10^{-4} s^{-1}. The elastic modulus was fitted using the first 0.5% strain. [b]Average fracture strain of polymer thin films as determined by film-on-water experiments. [c]Crack-onset strain of polymer film measured on 20:1 PDMS. [d]Thermally annealed at 150 °C for 30 min under nitrogen.

after curing that an enhancement in thin film ductility was observed.

It is however noteworthy that the COS was improved for the pristine polymer-cross-linker blends even prior to thermal curing. Also, the FOW measured fracture strain values were significantly lower than the COS from optical measurements, even though a similar trend was obtained. The enhanced COS for the film on elastomer test method is likely caused by stress delocalization along the support PDMS substrate, while in the FOW case, the stress is concentrated on film defects, which led to rapid crack propagation. Additionally, this may be due to the cross-linkers covalently binding to the elastomer for enhanced adhesion for the COS samples. FOW measurements on the other hand did not have any thin film polymer-elastomer interface with available sp^3 C−H groups for cross-linking; therefore, no improvement in fracture strain was observed for the pristine films. Furthermore, FOW samples were spin-coated on a layer of water-soluble dextran that is rougher than SiO_2 substrates and hence introduced more defects.

To assess the morphological changes of the polymer films upon strain, their dichroic ratio R was evaluated; it was defined as the UV−vis peak absorbance (A) of the film with polarized light parallel to the strain direction divided by that in the perpendicular direction ($R = A_{\parallel}/A_{\perp}$). An increase in R typically implies polymer chain alignment in the strain direction because the internal charge transfer peak has a transition dipole moment along the backbone direction.[44] As shown in Figure 9c, R increased initially for the neat semiconductor but started to plateau after ε = 25%. This was due to the formation of cracks that began to appear at ε = 25% according to COS measurements. The formation of cracks dissipates the energy upon strain, allowing the polymer film to relax and cease further alignment. Figure 9b clearly shows the AFM images of the large cracks formed at ε = 50 and 100%. On the contrary, the cross-linked films showed a steady linear increase in R with strains of ≤100%. A maximum R of 2.9 was achieved by **5Branch** at ε = 100%. The polymer crystalline alignment was further confirmed by cross-polarized images that displayed clear birefringence (Figure S16) and GIXD that showed a clear (010) $\pi-\pi$ diffraction peak in the parallel to strain direction but none in the perpendicular direction (Figure S13). The

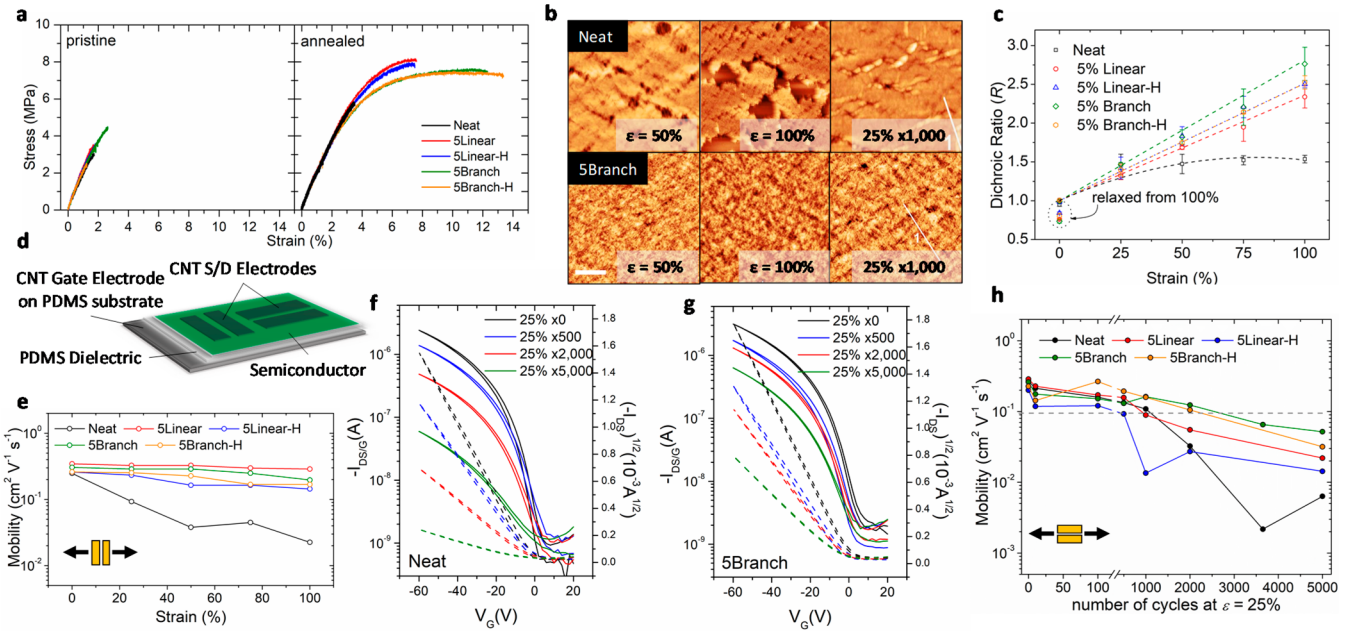


Figure 9. Mechanical and electric properties of cross-linked polymer thin films. (a) Stress−strain curve of pristine polymer-cross-linker blends (left) and cured polymer films obtained by the film-on-water (FOW) technique. Each sample was tested at least five times, and the curve with the maximum fracture strain is displayed. (b) Atom force microscope (AFM) images of neat semiconductor (top) and **5Branch** (bottom) films at ε = 50, 100, and 0% after 1000 cycles of 25% strain. The scale bar is 1 μm. (c) Dichroic ratio (R) of cross-linked films under different strains. (d) Schematic representation showing the device structure of the fully stretchable organic field-effect transistor (OFET) with PDMS dielectric and carbon nanotube electrodes. (e) Average mobility of strained polymer films in the parallel to strain direction. The mobility in the perpendicular to strain direction can be found in the Supporting Information. Transfer curves of the relaxed (f) neat semiconductor and (g) **5Branch** after various stretching cycles of ε = 25%. (h) Average mobility of relaxed polymer films after various stretching cycles.

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steady continuous increase in the level of alignment arose from the enhanced ductility of the polymer films, which allowed polymer chains to align at increasing strain as opposed to forming cracks. As shown in Figure 9b, no cracks were observed by AFM up to ε = 100% for **5Branch**.

To evaluate the elasticity of the cross-linked films, the R values of the polymer films were remeasured at ε = 0% after relaxing from ε = 100%. A fully elastic film exhibits reversible deformation and hence should return to its original morphology with an R of 1. For the neat polymer film, $R \approx 1$ upon relaxation, which was expected as the neat film failed prior to yielding according to FOW measurement (Figure 9a). The polymer started to form cracks at ε = 25% prior to yielding, allowing the strained film to fully recover upon relaxation. The cross-linked films on the other hand yielded prior to fracture. R values of ≈0.75 were recorded for all the cross-linked films upon relaxation from ε = 100%, indicating chain alignment in the perpendicular to strain direction. This suggests in-plane wrinkling of polymer chains as a result of yielding.

2.2.3. Cross-Linked Film Electric Properties. Organic field-effect transistor (OFET) measurements were performed to evaluate the effect of cross-linker additives on the charge transport properties of the polymer film. Device performance data in the bottom-gate-top-contact (BGTC) configuration on both *n*-octadecyltrimethoxysilane (OTMS) and benzocylobutene (BCB) modified SiO_2 substrates are summarized in Table 4. OTMS devices typically exhibit superior mobility but also non-ideal transfer with double-slope behavior (Figure S18).[45] A decrease in the field-effect hole mobility (μ_h) from 1.27 to 0.48 $cm^2\ V^{-1}\ s^{-1}$ along with a significant shift in the threshold voltage (V_{th}) from 14.2 to −1.9 V was recorded for OTMS devices upon cross-linking. Interestingly, for BCB devices, an average μ_h of ≈0.2 $cm^2\ V^{-1}\ s^{-1}$ and a V_{th} close to 0 V were measured for both neat and cross-linked films (Figure S19), suggesting the observed behavior of OTMS devices might be affected by traps at the active layer-substrate interface more than by possible doping interactions with the cross-linker additives. BCB devices further demonstrated that the polymer films could be readily cross-linked without the significant degradation of charge transport properties. Notably, **5inear-H** exhibited a lower μ_h of 0.14 $cm^2\ V^{-1}\ s^{-1}$ due to the film roughness as a result of cross-linker recrystallization.

Table 4. OFET Device Characteristics of Cross-Linked Films in the BGTC Configuration[a]

		μ_h ($cm^2\ V^{-1}\ s^{-1}$)[b]			
	substrate	avg	max	I_{on}/I_{off}[c]	V_{th} (V)[d]
neat	OTMS[e]	1.06	1.27	>10^5	14.2
	BCB[f]	0.24	0.27	>10^5	0.2
5Linear	OTMS	0.45	0.59	>10^5	−1.6
	BCB	0.19	0.20	>10^5	1.8
5Linear-H	OTMS	0.79	1.14	>10^5	0.6
	BCB	0.14	0.15	>10^5	1.6
5Branch	OTMS	0.39	0.48	>10^5	−0.8
	BCB	0.22	0.24	>10^5	−1.2
5Branch-H	OTMS	0.52	0.64	>10^5	−1.9
	BCB	0.18	0.22	>10^5	−4.1

[a]All polymer thin films were thermally annealed at 150 °C for 30 min under nitrogen before measurement. All values are averaged and extracted from at least six devices. The source-drain voltage was set to be −60 V for all measurements (L = 50 μm; W = 1000 μm). [b]Field-effect hole mobility. [c]On:off ratio of the device. [d]Threshold voltage of the device. [e]*n*-Octadecyltrimethoxysilane monolayer on 300 nm SiO_2 substrates. [f]Benzocyclobutene layer on 300 nm SiO_2 substrates.

To assess the charge transport properties of the polymer films under strain, fully stretchable BGTC OFETs were fabricated following previous procedures[23,24,37] and subjected to monotonic loading (Figure 9d). PDMS was used as the polymer dielectric and device substrate. Conductive carbon nanotubes (CNTs) were used as both source/drain and gate electrodes. Each layer was either directly spin-coated or spray-coated. Only a single transfer was required in the process, leading to smooth interfaces between the layers. A PDMS-based tough elastomer that we developed was used to embed the gate CNT network to prevent damage to the soft dielectric during cyclic loading.[46] The devices showed ideal transfer characteristics with low hysteresis (Figure 9f,g). An initial μ_h of ∼0.25 $cm^2\ V^{-1}\ s^{-1}$ was achieved for all polymer films comparable to BCB devices. The detailed fabrication procedure for the fully stretchable devices is available in the Supporting Information. As shown in Figure 9e, the μ_h of the neat polymer decreased with an increase in strain. The cross-linked films on the other hand displayed stable μ_h values of >0.2 $cm^2\ V^{-1}\ s^{-1}$ up to ε = 100% in the parallel to strain direction. Interestingly, μ_h in the direction perpendicular to strain decreases significantly to values comparable to that of the neat polymer film (Figure S21). This phenomenon could be explained by the strain-induced polymer alignment investigated earlier by R measurement and polarized optical microscopy. As the polymer chains aligned in the strain direction, charge transport in the direction perpendicular to strain occurs predominantly via interchain hopping leading to a lower μ_h.[44,47] Also, Poisson's effect leads to buckling in the perpendicular to strain direction, which could be another factor leading to a lower μ_h.

The straining of the stretchable OFETs could lead to thinning of the polymer dielectric as well as distortion of the CNT source/drain dimensions that would affect μ_h measurements. The correct capacitance and channel dimensions at different strains were therefore measured using Poisson's ratio (Table S1) and used to correctly extract μ_h values. To confirm the observed μ_h trends without the complications of a distorted dielectric layer and source/drain electrodes, rigid OFETs were fabricated. Semiconductor films were first strained on PDMS substrates and then transferred onto a BCB modified SiO_2 substrate following previous methods.[13,18,21] Au electrodes were subsequently evaporated on the strained films to give BGTC OFETs. As displayed in Figure S20, a similar trend with cross-linked films showing stable μ_h values in the parallel to strain direction was observed.

So far, we have investigated only the cross-linked films' ductility. As pointed out in our previous study, one major advantage of cross-linking semiconductor films is the improved elasticity leading to better device cyclability. The stretchable device of each polymer film was subjected to various cycles of ε = 25% for ≤5000 cycles and evaluated. Figure 9h summarizes the average μ_h of each polymer film at ε = 0% after different numbers of cycles in the perpendicular to strain direction. μ_h values in the parallel to strain direction could be found in the Supporting Information. Contrary to what was observed for the strained devices, the direction perpendicular to strain consistently gave higher μ_h values for the relaxed films (Figure 9h). This may be due to the partial alignment in the

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perpendicular direction upon relaxation as shown in dichroic ratio measurements. These devices were measured at a relaxed state that has wrinkling from plastic deformation. A similar trend was observed in our previous study that showed large wrinkles in the parallel to strain direction by AFM and hence a compromise in μ_h.[31] No obvious wrinkling was detected by AFM in this study (Figure 9b), but dichroic ratio measurements certainly suggest in-plane wrinkling due to yielding.

Focusing on the μ_h in the perpendicular direction, we found the cross-linked and neat polymer films showed comparable performance up to 500 cycles with a minimal drop in μ_h. However, at 1000 cycles, **5Linear-H** began to show compromised performance dropping to a μ_h of ≈0.01 cm^2 V^{-1} s^{-1}. Similarly, for the neat film, the device began to degrade at 2000 cycles. Surprisingly, despite showing a higher ductility, **5Linear-H** was less robust than the neat film. This may be due to its high surface roughness, which led to easy delamination between device layers as well as strain localization in thinner regions leading to local fracture. The neat polymer film had smooth interfaces but started to form cracks due to fatigue after 1000 cycles. Figure 9b shows the AFM image of a transferred neat polymer film after 1000 cycles of ε = 25%, which clearly showed cracks and wrinkles leading to μ_h decay. **5Linear**, **5Branch**, and **5Branch-H** on the other hand exhibited stable μ_h values for ≤5000 cycles of ε = 25%. **5Branch** and **5Branch-H** were clearly more robust than **5Linear**, with μ_h values of >0.1 cm^2 V^{-1} s^{-1} at 2000 cycles. Panels f and g of Figure 9 show the evolution of the transfer curves of the neat polymer and **5Branch** with an increasing number of cycles of ε = 25%. The good cyclability of **5Branch** is apparent, showing a slower decay of charge transport.

3. CONCLUSION

In summary, we synthesized four perfluorophenyl azide (PFPA) alkyl cross-linkers with different amounts of branching and H-bonding units and revealed the importance of cross-linker crystallinity in tuning polymer semiconductor thin film morphology. GIXD and DSC of the cross-linkers showed that the insertion of amide bonds could promote cross-linker crystallization by H-bonding, whereas more highly branched tertiary substitution could suppress crystalline packing. ATR-FTIR studies revealed that while both **Linear-H** and **Branch-H** contained H-bonded amide groups, **Linear-H** was in a crystalline state whereas **Branch-H** remained in an amorphous state. The high crystallinity of **Linear-H** led to the spontaneous phase separation and recrystallization of the cross-linker upon mixing with polymer semiconductor **DPP-10C$_5$DE**, as revealed by DSC, GIXD, and ATR-FTIR experiments. Some recrystallization was similarly detected by DSC for the **5Linear** cross-linker-polymer blend. Driven by their difference in crystallinity, two uniquely different cross-linked film morphologies were obtained as confirmed by AFM and optical microscopy. UV−vis and GIXD revealed a suppression of polymer aggregation and crystallization upon cross-linking for all cross-linkers and showed that they resided in the amorphous regions of the polymer film as the insertion of cross-linkers in the polymer crystalline region was highly unlikely. Because of the initial phase separation of pure crystalline cross-linker domains for **Linear** and **Linear-H** in the pristine cross-linker-polymer blend, upon thermal curing, the crystalline additives melted and cross-linked neighboring polymer chains to form concentrated cross-linked domains. This led to an uneven distribution of stress during loading and hence limited ductility as measured by COS and FOW experiments. The rough surface of **5Linear-H** further led to the poor adhesion of stretchable OFET and therefore low cyclability. **Branch** and **Branch-H** on the other hand, as a result of their low crystallinity and excellent miscibility with the neat semiconductor, exhibited smooth and evenly cross-linked film morphologies. The evenly cross-linked network allowed stress to be evenly distributed throughout the film, which led to greater device ductility and cyclability. Aside from the importance of cross-linker crystallinity to thin film deformability, this work further illustrates the complexity of introducing an energy dissipation mechanism through H-bonding units in polymer films. Despite the amorphous H-bonded amide peaks in both **5Linear-H** and **5Branch-H** films revealed by ATR-FTIR, no significant change in mechanical properties was observed possibly due to the strength and location of the H-bonds.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.chemmater.8b04314.

Synthesis, NMR, TGA, UV−vis, DSC, DMA, GIXD, optical images, and AFM images (PDF)

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Notes

The authors declare no competing financial interest.

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